

Lab 2b - ReadMe

Build and Run:

On almighty:

```
git clone <repo-url> (or just upload via FileZilla)
cd raytracer
mkdir build && cd build
cmake .. && make
```

Or with g++:

```
g++ -std=c++17 -O2 -Isrc -lexternal \ src/main.cpp src/parser/SceneParser.cpp
external/tinyxml2/tinyxml2.cpp \ -o raytracer
```

Running:

```
./raytracer scenes/example1.xml
./raytracer scenes/example2.xml
./raytracer scenes/example3.xml
./raytracer scenes/example4.xml
./raytracer scenes/example5.xml
./raytracer scenes/example6.xml
./raytracer scenes/example7.xml
./raytracer scenes/final_scene.xml
```

The output image is in .png, so in order to see the rendered image you have to change it to .ppm manually.

Implemented tasks:

AABBs,
SAH,
BVH,
EF8 - Spot Light
EF3 - Anti-aliased textures
final_scene.xml:
- Night Gallery scene with three pedestals with spheres

- Gold reflective sphere, glass refractive sphere, blue diffuse sphere
- Marble textured floor, brick textured back wall
- Three spot lights with distinct colors and falloff
- Demonstrates BVH, spot lights, bilinear texture filtering, reflection and refraction

Remarks:

The XML parser uses `tinyxml2`, vendored in `external/tinyxml2/`. No installation is required - it is compiled directly as part of the build.

Texture loading uses `stb_image.h`, vendored in `src/stb_image.h`. No installation required.